

Input Icons for UI Toolkit

An extension for Input Icons for TMPPro

Requirements

Before installing this extension, ensure you have:

- Unity 2021.2 or higher
- Input Icons for TMPPro (available on Asset Store: <https://u3d.as/2Lkz>)
- TextMeshPro 2.1.6 or higher
- Input System 1.2.0 or higher (1.4.4+ recommended)
- UI Toolkit (automatically included in Unity 2021+)

What is Input Icons for UI Toolkit?

This extension brings Input Icons functionality to Unity's modern UI Toolkit system. While the main Input Icons package works with Unity's legacy UI system (uGUI) and TextMeshPro, this extension enables you to display dynamic input prompts in UI Toolkit interfaces.

Once configured, you can display input bindings in UI Documents using style tags (like `<style=gameplay/jump>`) or visual components, with automatic updates when users switch between keyboard and controller.

Important Prerequisites

You must complete the main Input Icons for TMPPro setup before using this extension. This creates the necessary sprite atlases that the UI Toolkit extension converts for use in UI Toolkit interfaces.

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Updating From An Earlier Version

When you update from an earlier version, the references on the manager might get reset. So if you made any changes to the default Manager, Panel Settings, Text Settings, Theme Style or the Text Style Sheet asset, those changes will probably be lost and you will have to make your changes again. For that reason, you might want to create your own assets and link reference them on the manager so you don't have to do a huge amount of work every time you update this asset.

Also when you update, you will likely have to do the setup again in "Tools -> Input Icons -> Input Icons UI Toolkit Setup".

Getting Started

Step 1: Open the Setup Window

Navigate to **Tools → Input Icons → Input Icons UI Toolkit Setup** to open the streamlined setup window.

The setup is organized into four clear steps:

1. **Prerequisites** - Verify dependencies and main setup
2. **UI Toolkit Configuration** - Link UI Toolkit assets
3. **Sprite Asset Creation** - Generate UI Toolkit-compatible assets
4. **Testing & Validation** - Verify functionality

Step 2: Prerequisites Check

The setup window automatically validates:

- Input Icons Manager is assigned
- Icon Set Configurator is configured
- Sprite assets from main setup exist
- UI Toolkit Manager is created

If any prerequisites fail, the window provides direct links to resolve issues.

Update Setting: Choose how UI Documents refresh when input devices change:

- **Search and Update:** Automatically finds all UI Documents (recommended)
- **Via Components:** Requires II_UIDocument component (more performant)

Step 3: UI Toolkit Configuration

Configure the essential UI Toolkit assets. The **Text Settings** and **Text Style Sheet** (highlighted in green) are the most critical:

- **Theme Style Sheet:** UI Toolkit visual styling
- **Panel Settings:** References Text Settings
- **Text Settings:** Must reference Text Style Sheet

- **Text Style Sheet:** Contains input action styles

Important: Your Panel Settings must reference Text Settings, and Text Settings must reference the Text Style Sheet for icons to display in text elements.

Use the **Configure Assets** button to automatically link these assets together.

Adding Input Action Styles

Choose between two setup methods:

Quick Setup (Recommended):

1. Click "Prepare Style Sheet"
2. Wait for compilation
3. Click "Add Style Entries"

Manual Setup (Advanced):

1. Click "Prepare Style Sheet (Manual)"
2. Make a small change in Text Style Sheet inspector and undo it
3. Click "Add Style Entries (Manual)"

Step 4: Sprite Asset Creation

This step converts your TMPro sprite atlases into UI Toolkit-compatible Text Sprite Assets:

1. **Create Text Sprite Assets:** Generates assets from existing sprite atlases
2. **Auto-Assign Created Assets:** Attempts automatic assignment by name matching
3. **Organize Assets:** Moves and renames assets to proper locations
4. **Optimize Glyph Positions:** Improves icon alignment in text

The setup window provides separate fields for each controller type (Keyboard, PlayStation, Xbox, Nintendo Switch).

Step 5: Testing & Validation

The final step verifies your setup and provides usage instructions. A complete setup shows green checkmarks for:

- UI Toolkit Manager assigned
- UI Toolkit Assets configured
- Text Sprite Assets created

Displaying Input Icons

Method 1: In Text Elements (Style Tags)

Use style tags in any UI Toolkit text element with "Enable Rich Text" enabled:

Press <style=gameplay/jump> to jump

Press <style=gameplay/move> to move

Use <style=gameplay/move/up> for composite bindings

Format: <style=ActionMapName/ActionName> or <style=ActionMapName/ActionName/PartName>

Method 2: On Visual Elements (Components)

1. Give your visual elements unique names/IDs in UI Builder
2. Add **II_UITKImagePromptBehaviour** component to the GameObject with the UI Document
3. Configure the component to target specific visual elements by their IDs
4. The component will display input icons as background images on those elements

Asset Management

When updating the extension, your custom assets may be reset. Create your own UI Toolkit assets and reference them in the manager to preserve settings across updates.

Limitations

This extension shares limitations with the main Input Icons package:

- **Keyboard Support:** QWERTY, QWERTZ, AZERTY layouts (falls back to QWERTY for unsupported layouts)
- **Action Asset Conflicts:** Avoid multiple Input Action Assets with identical ActionMap/Action names
- **No Built-in Rebinding:** Use the main package's rebinding prefabs or create custom rebinding logic

Troubleshooting

Icons not displaying in text:

- Verify Text Settings references Text Style Sheet
- Check that sprite assets are assigned
- Ensure style names match Input Action names exactly

Component not working:

- Confirm `II_UITKImagePromptBehaviour` is on `GameObject` with UI Document
- Verify visual element IDs match component configuration
- Check that visual elements exist in the UI Document

Setup errors:

- Complete main Input Icons for TMPPro setup first
- Check console for missing reference errors
- Try recreating Text Sprite Assets

Assets missing after update:

- Redo setup process
- Create custom asset copies to preserve across updates

Advanced Features

Custom Asset Setup

You can create and use your own UI Toolkit assets instead of the provided examples. The key requirement is maintaining the relationship: Panel Settings → Text Settings → Text Style Sheet.

Performance Optimization

For better performance in projects with many UI Documents, use the "Via Components" update method and add `II_UIDocument` components only to UI Documents that display Input Icons.

Style Sheet Management

Use the "Remove All Input Icon Styles" button in the Help section to clean the Text Style Sheet if you need to start over or remove Input Icons integration.

Support

For technical support or questions:

- Email: tobias.froihofer@gmx.at
- Check the main Input Icons documentation
- Review Unity console for error messages
- Try the troubleshooting steps above

Before contacting support, please verify you've completed the main Input Icons setup and followed all prerequisite steps in the UI Toolkit setup window.